



Universidad autónoma del Estado de
México

Ingeniería en computación

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Sistemas operativos

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1. BTOP

btop es una herramienta de monitorización del sistema en tiempo real y altamente personalizable que ofrece a los usuarios una interfaz intuitiva y visualmente atractiva para monitorizar los recursos del sistema.

Ofrece una descripción general completa del rendimiento de su sistema que incluye el uso de la CPU, el consumo de memoria, la actividad del disco, el ancho de banda de la red y los procesos que se ejecutan en su sistema.

btop: La herramienta definitiva de monitoreo del sistema en tiempo real. (s.f.). Artículos actuales sobre Linux: programas, distribuciones, instrucciones detalladas. <https://es.linux-console.net/?p=35623>

The first screenshot shows the terminal output of the command `sudo apt install btop`. The output indicates that btop is already installed and that the newest version (1.3.0-1) is available. The user is prompted to confirm the installation, and they respond with 'Y'.

```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ sudo apt install btop  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
btop is already the newest version (1.3.0-1).  
0 upgraded, 0 newly installed, 0 to remove and 165 not upgraded.  
6 not fully installed or removed.  
After this operation, 0 B of additional disk space will be used.  
Do you want to continue? [Y/n]
```

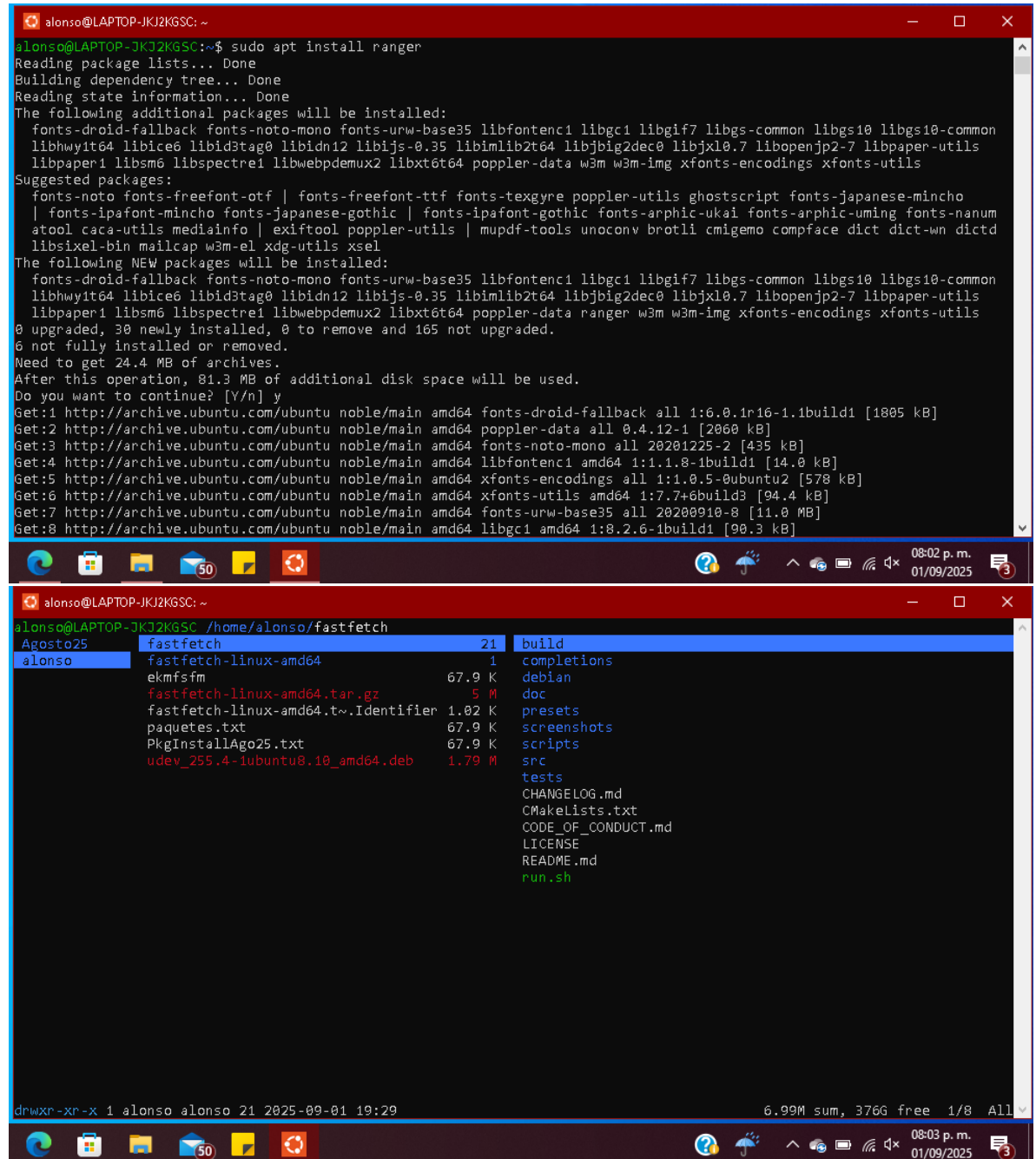
The second screenshot shows the btop interface running in the terminal. The interface displays various system metrics, including CPU usage, memory usage, disk usage, and network activity. The CPU usage is shown as 15% for the N3060 processor. The memory usage is shown as 3.84 GiB total, with 2.69 GiB used and 1.14 GiB available. The disk usage is shown as 12.0 GiB total, with 160 MiB used and 11.8 GiB free. The network activity is shown as 0 Byte/s download and 0 Byte/s upload. The interface also displays a list of running processes, including btop, init, and bash.

```
alonso@LAPTOP-JKJ2KGSC: ~  
1cpu menu preset * 19:57:05 BAT 99% 2000ms  
up 02:01:45  
2mem disks io 4proc filter per-core reverse tree <cpu lazy >  
Total: 3.84 GiB swap 12.0 GiB  
Used: 2.69 GiB Used: 1% 160 MiB  
Available: 1.14 GiB Free: 99% 11.8 GiB  
Cached: 184 MiB  
Free: 993 MiB  
3net sync auto zero cb eth0 n  
download  
0 Byte/s (0 bits)  
Total: 0 Byte  
upload  
0 Byte/s (0 bits)  
Total: 0 Byte  
Pid: Program: Command: User: MemB Cpu%  
5245 btop btop alon+ 3.8M 0.4  
1 init /init root 660K 0.0  
116 bash -bash alon+ 3.8M 0.0  
115 init /init root 228K 0.0  
↑ select ↓ info @ terminate U kill U signals 0/4
```

2. RANGER

Ranger es un potente y uno de los administradores de archivos CLI más utilizados de Linux. Es una herramienta liviana y fácil de usar con varias funciones para navegar y administrar los archivos en la terminal. El programa puede realizar tareas de administración de archivos utilizando unas cuantas teclas o la entrada del ratón. Este gestor de archivos también puede ser configurado para abrir archivos con programas predefinidos y para mostrar una vista previa del archivo seleccionado llamando a los programas externos. *Ranger, un gestor de archivo ligero para la Terminal de Ubuntu.* (s.f.).

Ubunlog. <https://ubunlog.com/ranger-gestor-archivo-terminal/>

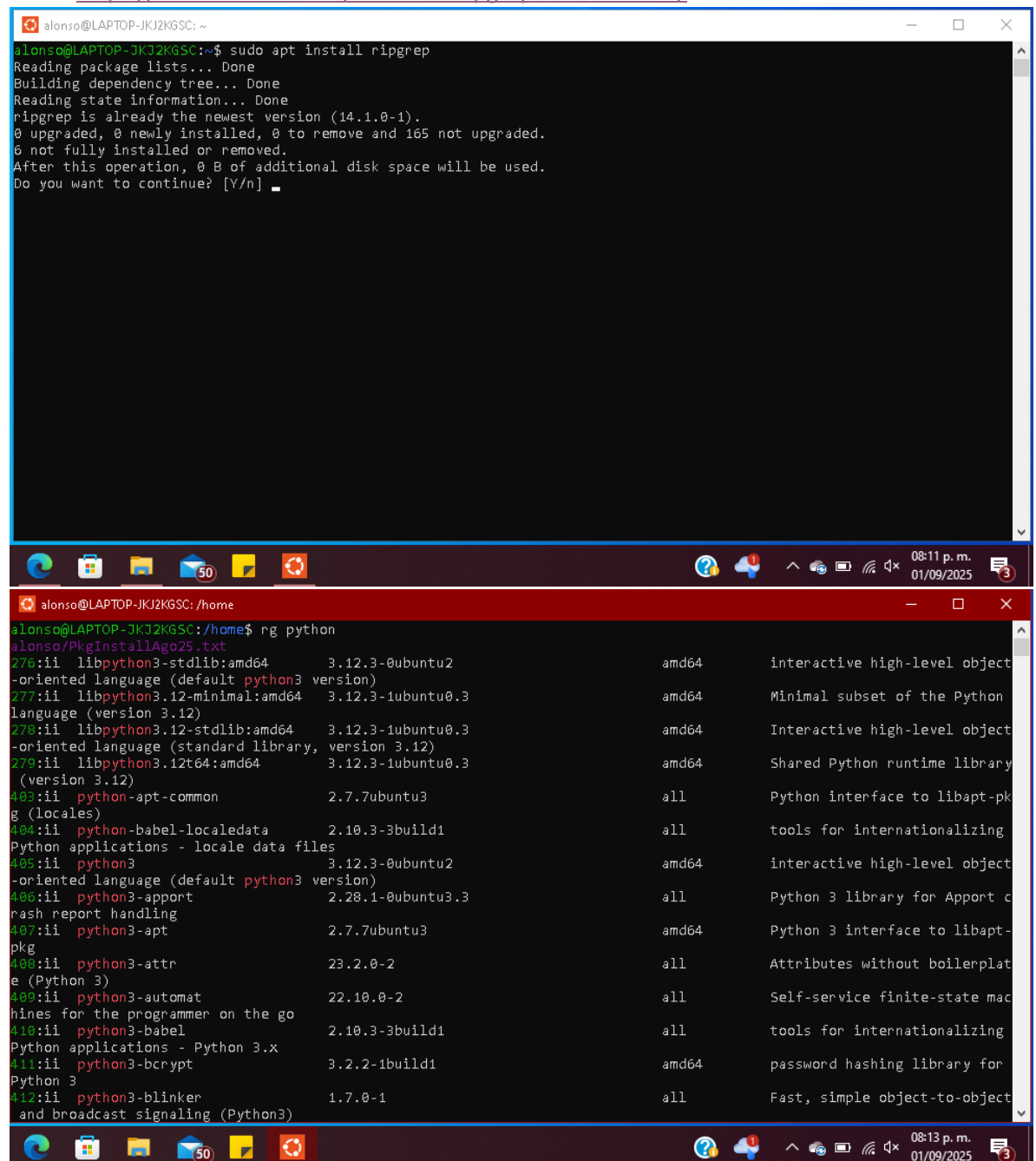


```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ sudo apt install ranger  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  fonts-droid-fallback fonts-noto-mono fonts-urw-base35 libfontenc1 libgc1 libgif7 libgs-common libgs10 libgs10-common  
  libhwy1t64 libice6 libid3tag0 libidn12 libijs-0.35 libimlib2t64 libjbig2dec0 libjxl0.7 libopenjp2-7 libpaper-utils  
  libpaper1 libsm6 libspectre1 libwebpdemux2 libxt6t64 poppler-data w3m w3m-img xfonts-encodings xfonts-utils  
Suggested packages:  
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf fonts-texgyre poppler-utils ghostscript fonts-japanese-mincho  
  | fonts-ipafont-mincho fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming fonts-nanum  
  atool caca-utils mediainfo | exiftool poppler-utils | mupdf-tools unoconv brotli cmigemo compface dict dict-wn dictd  
  libixel-bin mailcap w3m-el xdg-utils xsel  
The following NEW packages will be installed:  
  fonts-droid-fallback fonts-noto-mono fonts-urw-base35 libfontenc1 libgc1 libgif7 libgs-common libgs10 libgs10-common  
  libhwy1t64 libice6 libid3tag0 libidn12 libijs-0.35 libimlib2t64 libjbig2dec0 libjxl0.7 libopenjp2-7 libpaper-utils  
  libpaper1 libsm6 libspectre1 libwebpdemux2 libxt6t64 poppler-data ranger w3m w3m-img xfonts-encodings xfonts-utils  
0 upgraded, 30 newly installed, 0 to remove and 165 not upgraded.  
Need to get 24.4 MB of archives.  
After this operation, 81.3 MB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1805 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 poppler-data all 0.4.12-1 [2060 kB]  
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-noto-mono all 20201225-2 [435 kB]  
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libfontenc1 amd64 1:1.1.8-1build1 [14.0 kB]  
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 xfonts-encodings all 1:1.0.5-0ubuntu2 [578 kB]  
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 xfonts-utils amd64 1:7.7+6build3 [94.4 kB]  
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-urw-base35 all 20200910-8 [11.0 MB]  
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 libgc1 amd64 1:8.2.6-1build1 [90.3 kB]  
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC /home/alonso/fastfetch  
Agosto25 fastfetch 21 build  
alonso fastfetch-linux-amd64 1 completions  
ekmfsfm 67.9 K debian  
fastfetch-linux-amd64.tar.gz 5 M doc  
fastfetch-linux-amd64.t~.Identifier 1.02 K presets  
paquetes.txt 67.9 K screenshots  
PkgInstallAgo25.txt 67.9 K scripts  
udev_255.4-1ubuntu8.10_amd64.deb 1.79 M src  
tests  
CHANGELOG.md  
CMakeLists.txt  
CODE_OF_CONDUCT.md  
LICENSE  
README.md  
run.sh  
drwxr-xr-x 1 alonso alonso 21 2025-09-01 19:29 6.99M sum, 376G free 1/8 All
```

3. RIPGREP

ripgrep, abreviado como “rg”, es una poderosa herramienta de búsqueda de línea de comandos que destaca por su velocidad y eficiencia. Está diseñada para buscar recursivamente directorios con un patrón de expresión regular, lo que la convierte en una herramienta para desarrolladores, administradores de sistemas y cualquier persona que necesite examinar grandes bases de código o archivos de texto. ripgrep ofrece funciones únicas como respetar tus archivos .gitignore e .ignore, lo que proporciona una experiencia de búsqueda más personalizada.

Cómo Utilizar el Comando ripgrep en Linux » Blog Linux. (s.f.). Adictos a Linux. <https://adictosalinux.com/comando-ripgrep-como-utilizar/>



The image shows a Linux desktop environment with two windows. The top window is a terminal titled 'alonso@LAPTOP-JKJ2KGSC: ~' showing the command 'sudo apt install ripgrep' and its output. The bottom window is a file manager titled 'alonso@LAPTOP-JKJ2KGSC: /home' showing the output of the command 'rg python' in the directory '/home/PkgInstallAgo25.txt'.

```
alonso@LAPTOP-JKJ2KGSC: ~  
alonso@LAPTOP-JKJ2KGSC:~$ sudo apt install ripgrep  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
ripgrep is already the newest version (14.1.0-1).  
0 upgraded, 0 newly installed, 0 to remove and 165 not upgraded.  
6 not fully installed or removed.  
After this operation, 0 B of additional disk space will be used.  
Do you want to continue? [Y/n] _  
  
alonso@LAPTOP-JKJ2KGSC: /home$ rg python  
alonso/PkgInstallAgo25.txt  
276:ii libpython3-stdlib:amd64 3.12.3-0ubuntu2 amd64 interactive high-level object  
-oriented language (default python3 version)  
277:ii libpython3.12-minimal:amd64 3.12.3-1ubuntu0.3 amd64 Minimal subset of the Python  
language (version 3.12)  
278:ii libpython3.12-stdlib:amd64 3.12.3-1ubuntu0.3 amd64 Interactive high-level object  
-oriented language (standard library, version 3.12)  
279:ii libpython3.12t64:amd64 3.12.3-1ubuntu0.3 amd64 Shared Python runtime library  
(version 3.12)  
403:ii python-apt-common 2.7.7ubuntu3 all Python interface to libapt-pk  
g (locales)  
404:ii python-babel-localedata 2.10.3-3build1 all tools for internationalizing  
Python applications - locale data files  
405:ii python3 3.12.3-0ubuntu2 amd64 interactive high-level object  
-oriented language (default python3 version)  
406:ii python3-apport 2.28.1-0ubuntu3.3 all Python 3 library for Apport c  
rash report handling  
407:ii python3-apt 2.7.7ubuntu3 amd64 Python 3 interface to libapt-  
pkg  
408:ii python3-attr 23.2.0-2 all Attributes without boilerplat  
e (Python 3)  
409:ii python3-automat 22.10.0-2 all Self-service finite-state mac  
hines for the programmer on the go  
410:ii python3-babel 2.10.3-3build1 all tools for internationalizing  
Python applications - Python 3.x  
411:ii python3-bcrypt 3.2.2-1build1 amd64 password hashing library for  
Python 3  
412:ii python3-blinker 1.7.0-1 all Fast, simple object-to-object  
and broadcast signaling (Python3)
```